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$$\begin{aligned} & \lim_{x \rightarrow +\infty} \left(x - x^2 \ln \left(1 + \frac{1}{x} \right) \right) \\ \Leftrightarrow & \lim_{x \rightarrow +\infty} \left(1 - 2x \ln \left(1 + \frac{1}{x} \right) + x^2 \cdot \frac{1}{1 + \frac{1}{x}} \cdot \frac{x \cdot 0 - 1 \cdot 1}{x^2} \right) \\ \Leftrightarrow & \lim_{x \rightarrow +\infty} \left(1 - 2x \ln \left(1 + \frac{1}{x} \right) + \frac{1}{1 + \frac{1}{x}} \right) \end{aligned}$$

$$\text{Stel } t = \frac{1}{x} \Rightarrow x = \frac{1}{t}$$

Wanneer $x \rightarrow \infty$, geldt dat $t \rightarrow 0$

We herschrijven de limiet in termen van t :

$$\begin{aligned} & \lim_{t \rightarrow 0} \left(\frac{1}{t} - \frac{1}{t^2} \ln(1+t) \right) \\ &= \lim_{t \rightarrow 0} \left(\frac{t - \ln(1+t)}{t^2} \right) \\ &= \lim_{t \rightarrow 0} \frac{1 - \frac{1}{1+t}}{2t} \\ &= \lim_{t \rightarrow 0} \frac{t}{2t + 2t^2} \\ &= \lim_{t \rightarrow 0} \frac{1}{2 + 2t} \\ &= \frac{1}{2} \end{aligned}$$

References